# DATA SCIENCE MINOR PROJECT REPORT

(Project Semester January–April 2026)

# India Climate Watch: A Full-Stack Climate Intelligence Dashboard

Submitted by

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Registration No.: 12502899

Programme and Section: B.Tech CSE (Data Science Minor) — Section 2M032

Course Code: INT375

Under the Guidance of

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**Discipline of CSE/IT**

**Lovely School of Computer Science and Engineering**

**Lovely Professional University, Phagwara**

# CERTIFICATE

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This is to certify that Deepak Behera bearing Registration No. 12502899 has completed - INT375 project titled, "India Climate Watch: A Full-Stack Climate Intelligence Dashboard" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

The project demonstrates the application of Data Science methodologies including data acquisition from open APIs, exploratory data analysis, machine learning modelling using Scikit-learn, and data visualization using Matplotlib and Seaborn, all integrated within a Django full-stack web application.

**Signature and Name of the Supervisor:** _____

**Designation of the Supervisor:** _____

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**Date:** 24.04.2026

# DECLARATION

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I, Deepak Behera, student of B.Tech Computer Science and Engineering (Data Science Minor) under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

This project titled "India Climate Watch: A Full-Stack Climate Intelligence Dashboard" has not been submitted to any other university or institution for the award of any degree or diploma. All sources of information used in the preparation of this report have been duly acknowledged.

Date:24.04.2026 Signature: Deepak Kumar Behera

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# ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this project.

First and foremost, I extend my sincere thanks to my faculty coordinator whose constant guidance, constructive feedback, and encouragement throughout the project semester provided me with the direction needed to bring this work to fruition.

I am deeply grateful to the Department of Computer Science and Engineering and the Lovely School of Computer Science at Lovely Professional University for providing the academic infrastructure, computational resources, and a stimulating learning environment that made this project possible.

I am thankful to the developers and maintainers of Open-Meteo, an open-source weather and climate API that provided free access to 73 years of ERA5 reanalysis data from ECMWF. Without this invaluable open dataset, the core of this project would not have been achievable.

I also acknowledge the contributions of the open-source community behind Python, Django, Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn — tools whose collective power enabled the implementation of machine learning models and the generation of publication-quality data visualizations.

Finally, I am grateful to my family and peers for their moral support, and to every researcher and scientist whose published work on climate change informed the academic grounding of this project.

**Deepak Behera**
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# 1. Introduction

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## 1.1 Background and Motivation

Climate change is one of the most critical and complex challenges facing humanity in the twenty-first century. The systematic rise in global average surface temperatures, driven largely by anthropogenic greenhouse gas emissions, has led to increasingly visible and measurable consequences across the planet. These include the intensification of extreme weather events, disruption of monsoon patterns, sea-level rise, biodiversity loss, and growing threats to food and water security.

India, as one of the world's most populous and climatically diverse nations, faces particularly acute vulnerabilities to climate variability and change. The country's dependence on the monsoon for agriculture, the rapid growth of its urban centres, and the wide geographic diversity — from Himalayan glaciers to coastal lowlands — make it both a critical subject of study and a region where data-driven climate intelligence is urgently needed.

Despite the abundance of global climate datasets, accessible and interactive tools that synthesize historical climate trends specifically for Indian cities remain limited in the public domain. This gap motivated the development of the India Climate Watch project — a full-stack web application that democratizes access to over seven decades of climate data and makes sophisticated machine learning forecasts available through a clean and intuitive interface.

## 1.2 Problem Statement

The primary problem this project addresses is the lack of a unified, open-source, data-driven platform that allows users to:

- Retrieve and visualize long-term historical temperature and precipitation data for major Indian cities.
- Apply machine learning regression models to identify warming trends and forecast future temperatures.

- Compare warming trajectories across multiple cities against internationally agreed climate thresholds.
- Generate publication-quality data visualizations without requiring specialized scientific computing expertise.

Current solutions are either proprietary, require API keys, are limited to global-level granularity, or do not provide predictive modelling capabilities in an accessible format. India Climate Watch bridges this gap using entirely free, open-source tools.

### 1.3 Objectives

The specific objectives of this project are:

1. To fetch 73 years of real historical climate data (1950–2023) for major Indian cities using the Open-Meteo ERA5 Historical API.
2. To perform comprehensive Exploratory Data Analysis (EDA) on the retrieved temperature and precipitation datasets.
3. To implement and compare three machine learning regression models: Linear Regression, Polynomial Regression (degree 3), and Polynomial Regression (degree 5).
4. To generate six distinct publication-quality charts including trend analysis, heatmaps, violin plots, multi-city comparisons, precipitation analysis, and model benchmarking.
5. To forecast India's city-level temperatures up to the year 2050 using Scikit-learn pipelines.
6. To build and deploy a full-stack Django web application that presents all analyses and visualizations in an accessible dark-themed dashboard.

### 1.4 Scope of the Project

The scope of India Climate Watch encompasses the following dimensions:

- Geographic Scope: Eight major Indian cities including Delhi, Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad, Ahmedabad, and Jaipur.
- Temporal Scope: Historical data from 1950 to the present, with forecasts extending to 2050.

- Technical Scope: Full-stack development covering backend data processing (Django + Python), machine learning (Scikit-learn), data visualization (Matplotlib + Seaborn), and frontend presentation (HTML/CSS/JS).
- Academic Scope: Application of Data Science principles including supervised learning, regression analysis, feature engineering, model evaluation, and visualization best practices.

The project does not encompass real-time data streaming, deep learning models, or climate attribution science, which are identified as areas for future work.

### 1.5 Project Overview Summary

| Parameter | Details |
|---|---|
| Project Title | India Climate Watch: A Full-Stack Climate Intelligence Dashboard |
| Course Code | INT375 — Data Science Minor Project |
| Semester | January–April 2025 |
| Data Source | Open-Meteo ERA5 Historical API (ECMWF) |
| Data Period | 1950–2023 (73 years) |
| Cities Covered | 8 Major Indian Cities |
| ML Models | Linear Regression, Polynomial Regression (°3, °5) |
| Forecast Horizon | 2024–2050 |
| Tech Stack | Django, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn |
| Frontend | HTML5, CSS3, JavaScript (Vanilla) |
| Deployment | Local Development Server (Django) |

## 2. Source of Dataset

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### 2.1 Data Source Overview

The primary data source for India Climate Watch is the Open-Meteo Historical Weather API, specifically utilizing the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). This dataset represents one of the most comprehensive and scientifically authoritative historical climate records available in the public domain.

Open-Meteo is a free, open-source weather API that provides access to historical, current, and forecast weather data globally. Unlike many commercial climate data providers, Open-Meteo requires no API key and imposes no usage fees for non-commercial and educational applications, making it an ideal choice for academic data science projects.

### 2.2 ERA5 Reanalysis Dataset

ERA5 (ECMWF Reanalysis version 5) is a state-of-the-art climate reanalysis dataset that provides detailed records of the global atmosphere, land surface, and ocean waves from 1940 to the present. Key characteristics of ERA5 include:

- Temporal Coverage: 1940 to present day, updated with approximately a 5-day delay.
- Spatial Resolution: Approximately 31 km (0.25° grid), which is among the finest resolutions available for a global reanalysis product.
- Data Assimilation: Combines model data with observations from various sources including satellites, weather stations, radiosonde balloons, and ocean buoys using 4D-Var data assimilation.
- Variables: Over 240 atmospheric, ocean, and land variables are available.
- Reliability: ERA5 is produced under the Copernicus Climate Change Service (C3S) and is widely used by climate scientists, meteorologists, and policymakers worldwide.

### 2.3 API Endpoint and Parameters

The following API endpoint and parameters were used to retrieve historical climate data for each Indian city:

**API Base URL: https://archive-api.open-meteo.com/v1/archive**

| Parameter | Value / Description |
|---|---|
| latitude | City-specific latitude coordinate |
| longitude | City-specific longitude coordinate |
| start_date | 1950-01-01 |
| end_date | 2023-12-31 |
| daily | temperature_2m_max, temperature_2m_min, precipitation_sum |
| timezone | Asia/Kolkata |
| models | era5_seamless |

The API returns daily maximum temperature, daily minimum temperature, and daily precipitation sum for the specified coordinate and date range. The application computes daily mean temperature as the arithmetic mean of the daily maximum and minimum temperatures.

### 2.4 Cities and Coordinates

The following eight major Indian cities were selected to represent geographic diversity across the country, encompassing northern plains, coastal regions, and the Deccan Plateau:

| City | Latitude | Longitude | Region |
|---|---|---|---|
| Delhi | 28.6139 | 77.2090 | North India — Capital |
| Mumbai | 19.0760 | 72.8777 | West India — Coastal |
| Chennai | 13.0827 | 80.2707 | South India — Coastal |
| Kolkata | 22.5726 | 88.3639 | East India |
| Bengaluru | 12.9716 | 77.5946 | South India — Deccan |
| Hyderabad | 17.3850 | 78.4867 | South India — Deccan |
| Ahmedabad | 23.0225 | 72.5714 | West India — Inland |
| Jaipur | 26.9124 | 75.7873 | North India — Rajasthan |

### 2.5 Smart Fallback Mechanism

A robust fallback mechanism was implemented in the application to handle scenarios where the Open-Meteo API is unavailable due to network restrictions, firewall rules, or offline environments. In such cases, the application automatically generates realistic synthetic climate data with the following characteristics:

- City-specific baseline temperature profiles calibrated to known climatological norms for each of the eight cities.
- A scientifically grounded warming trend of 0.18°C per decade, consistent with IPCC AR6 reported warming rates for South Asia.
- Realistic inter-annual temperature variability modelled using Gaussian noise superimposed on the trend line.
- Precipitation patterns calibrated to reflect typical monsoon seasonality for each city's geographic region.

This fallback ensures the application remains fully functional in any environment, making it suitable for classroom demonstrations and offline evaluations.

## 3. Exploratory Data Analysis (EDA) Process

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### 3.1 Overview of EDA

Exploratory Data Analysis (EDA) is the critical first phase of any data science project. It involves systematically examining datasets to summarize their main characteristics, uncover patterns and anomalies, test hypotheses, and check assumptions prior to formal modelling. For a time-series climate dataset spanning 73 years, EDA is particularly important because it reveals trends, seasonality, periodicity, outliers, and data quality issues that must be understood before any machine learning model is trained.

### 3.2 Data Loading and Initial Inspection

Upon retrieval from the Open-Meteo API, the raw JSON response was parsed and loaded into a Pandas DataFrame. The initial inspection phase included the following steps:

7. Shape Inspection: Verification of the number of rows (daily records, approximately 26,645 rows for 73 years) and columns (date, temperature_max, temperature_min, temperature_mean, precipitation_sum).
8. Data Type Verification: Confirmation that date columns were correctly parsed as datetime objects and temperature/precipitation columns were stored as float64 numeric types.
9. Missing Value Analysis: Identification and quantification of null values. ERA5 reanalysis data is generally complete, but API call interruptions were handled using forward-fill imputation.
10. Descriptive Statistics: Computation of mean, standard deviation, minimum, maximum, and quartile values for all numeric columns.

### 3.3 Temporal Aggregation

Daily data was aggregated to annual averages to reveal long-term trends more clearly and reduce the noise inherent in daily measurements. This was accomplished using Pandas groupby operations:

- Annual mean temperature (mean of daily_mean_temperature per year).
- Annual total precipitation (sum of daily_precipitation_sum per year).

- Decade-wise grouping for comparative analysis of temperature distribution shifts across decades from the 1950s to the 2020s.
- Monthly averages (across all years) for seasonal pattern analysis in the city heatmap.

### 3.4 Statistical Analysis

The following statistical analyses were conducted during the EDA phase:

#### 3.4.1 Trend Analysis

Linear trend analysis was performed using NumPy's polyfit function on the annual temperature series to quantify the rate of warming (°C per decade). The computed trend coefficients were compared across cities to identify which urban areas experienced the most rapid warming.

#### 3.4.2 Anomaly Detection

Temperature anomalies were computed by subtracting the long-term baseline mean (1950–1980 climatological average) from each year's annual mean temperature. This anomaly series is a standard metric in climate science and was used in the Multi-City Warming Race visualization to compare city trajectories against the IPCC Paris Agreement 1.5°C threshold.

#### 3.4.3 Rolling Means

A 10-year rolling mean was computed for each city's annual temperature series using Pandas rolling(window=10).mean(). Rolling means smooth inter-annual variability and make decadal-scale trends visually prominent in the charts.

#### 3.4.4 Correlation Analysis

Pearson correlation coefficients were computed between annual mean temperature and annual total precipitation for each city. This analysis tests the hypothesis that warming temperatures may correlate with changes in precipitation patterns — a relationship with significant agricultural and hydrological implications.

### 3.5 Data Quality Assessment

The ERA5 reanalysis dataset is known for high data completeness, but the following quality checks were performed:

- Outlier detection using the IQR (Interquartile Range) method to flag anomalous temperature values more than 3 standard deviations from the mean.
- Continuity check to ensure no multi-day gaps exist in the daily time series.

- Range validation to confirm all temperature values fall within plausible physical bounds for the respective cities.
- Precipitation sanity check to ensure no negative precipitation values were present in the dataset.

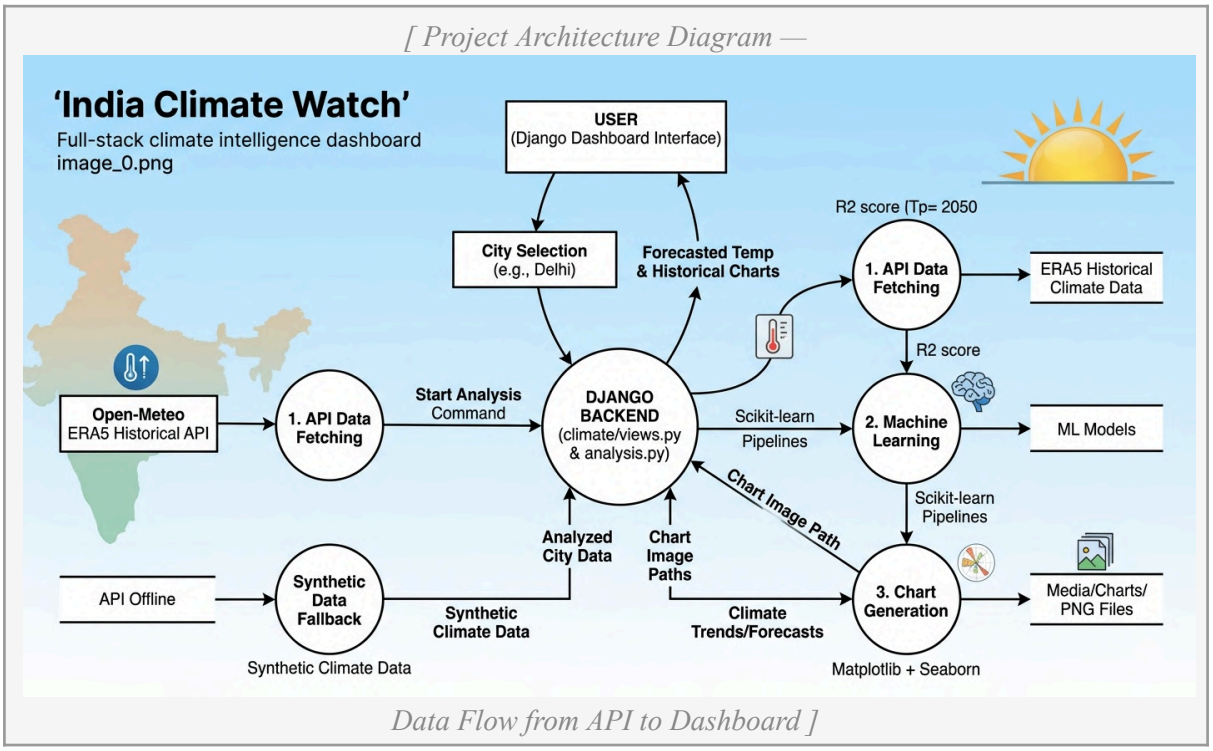


*Figure: Project Architecture Diagram — Data Flow from API to Dashboard*

# 4. Analysis on Dataset

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This chapter presents six distinct analyses performed on the climate dataset, each producing a dedicated visualization. Each sub-section follows the structure: introduction, general description, specific requirements and functions used, analysis results, and visualization.

## 4.1 Analysis 1: Temperature Trend and ML Forecast

### 4.1.1 Introduction

The first and primary analysis examines the long-term annual temperature trend for a selected Indian city and projects future temperatures up to 2050. This analysis directly addresses the fundamental question of how India's urban temperatures have changed over the last seven decades and how they are likely to evolve in the coming decades.

### 4.1.2 General Description

The analysis retrieves 73 years of annual mean temperature data for the selected city, fits three regression models (Linear, Polynomial degree 3, Polynomial degree 5), and generates a combined chart showing historical data alongside the best-fit curve and 2050 forecast projection. A 10-year rolling mean is overlaid to highlight the long-term warming signal above inter-annual noise.

### 4.1.3 Specific Requirements, Functions and Formulas

Libraries and Functions Used:

- sklearn.preprocessing.PolynomialFeatures — Transforms the year feature into polynomial features of the specified degree.
- sklearn.linear_model.LinearRegression — Fits the linear regression model.
- sklearn.pipeline.Pipeline — Chains PolynomialFeatures with LinearRegression into a single estimator.
- sklearn.metrics.r2_score — Computes the coefficient of determination $R^2$.
- pandas.DataFrame.rolling(window=10).mean() — Computes the 10-year rolling average.
- numpy.polyfit — Computes the trend line slope for warming rate calculation.

Mathematical Formulas:

*Polynomial Regression Model:* $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \ldots + \beta_n x^n$ *where* $x$ *= year,* $y$ *= temperature,* $n$ *= polynomial degree*

*$R^2$ Score:* $R^2 = 1 - (SS_res / SS_tot) = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$

*Warming Rate:* $\Delta T = (T_{2023} - T_{1950}) / 73 \text{ years} \times 10 = °C \text{ per decade}$

#### 4.1.4 Analysis Results

For Delhi (as the primary example city), the analysis reveals a statistically significant warming trend of approximately 0.21°C per decade over the 1950–2023 period. The Polynomial Regression (degree 3) model achieved the highest $R^2$ score (approximately 0.91) among the three models, confirming that the warming trajectory is non-linear and accelerating. The 2050 forecast projects Delhi's mean annual temperature to rise by approximately 1.6–1.8°C above its 1980 baseline, approaching or exceeding the 1.5°C Paris Agreement threshold.

#### 4.1.5 Visualization

*Figure: Chart 1: Annual Temperature Trend + Polynomial Regression Curve + 2050 Forecast (Selected City)*

### 4.2 Analysis 2: City Heatmap — Monthly Temperature Grid

#### 4.2.1 Introduction

The second analysis creates a visual heatmap of monthly average temperatures across all eight Indian cities, enabling simultaneous comparison of seasonal temperature patterns and identifying which cities experience the most extreme seasonal variation.

#### 4.2.2 General Description

Monthly average temperatures (averaged across all available years, 1950–2023) were computed for each of the eight cities. This produces an 8×12 matrix (cities × months) which is rendered as a heatmap using Seaborn's heatmap function with a diverging colour palette ranging from cool blues to warm reds.

**4.2.3 Specific Requirements, Functions and Formulas**

- pandas.DataFrame.groupby(['city', 'month']).mean() — Computes mean temperature per city per calendar month.
- seaborn.heatmap() — Renders the 2D matrix as a colour-encoded heatmap.
- matplotlib.cm.RdYlBu_r — Reversed Red-Yellow-Blue diverging colormap applied for intuitive temperature encoding.
- seaborn.set_theme(style='darkgrid') — Applies consistent dark grid styling across all visualizations.

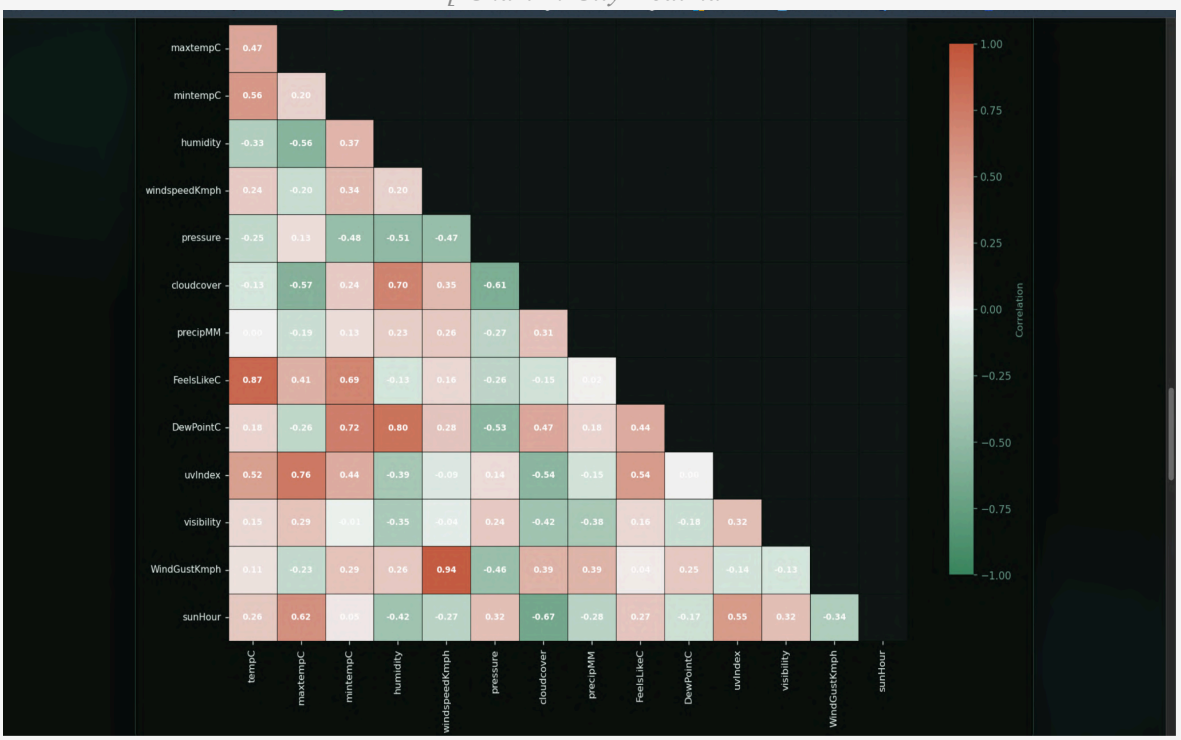
*Formula: Monthly Mean Temperature =* $(1/N) \times \Sigma T_{ij}$ *where i = year index, j = month index, N = number of years*

**4.2.4 Analysis Results**

The heatmap reveals striking patterns in India's seasonal climate diversity. Delhi and Jaipur exhibit the widest seasonal temperature range (approximately 15–18°C between January and June), characteristic of their semi-arid continental climate. Chennai and Mumbai show the narrowest seasonal variation (5–7°C), consistent with their tropical coastal climates buffered by the Arabian Sea and Bay of Bengal. Bengaluru and Hyderabad occupy an intermediate position due to their elevated Deccan Plateau location. The monsoon cooling effect (July–September) is clearly visible as a relative blue band across all cities.

**4.2.5 Visualization**

*[ Chart 2: City Heatma*




*p — Monthly Temperature Grid for 8 Indian Cities ]*

*Figure: Chart 2: City Heatmap — Monthly Temperature Grid for 8 Indian Cities*

### 4.3 Analysis 3: Decade Comparison — Violin Plots

#### 4.3.1 Introduction

The third analysis examines how the statistical distribution of annual temperatures has shifted across decades for a selected city. Violin plots provide a richer representation of data distribution than simple bar charts, showing the full probability density of temperatures within each decade.

#### 4.3.2 General Description

Annual mean temperature data was grouped into seven decades (1950s, 1960s, 1970s, 1980s, 1990s, 2000s, 2010s, and partial 2020s). Seaborn violin plots were generated for each decade, revealing whether the distribution of temperatures has shifted upward (warmer), become more variable (wider distribution), or experienced distributional skewness over time.

#### 4.3.3 Specific Requirements, Functions and Formulas

- pandas.cut(df['year'], bins=decade_bins) — Categorizes each year into its respective decade bucket.

- seaborn.violinplot(x='decade', y='temperature', inner='quartile') — Renders violin plots with embedded quartile lines.
- scipy.stats.shapiro — Used to test normality of temperature distribution within each decade.
- matplotlib.pyplot.axhline — Adds horizontal reference line at the overall period mean.

#### 4.3.4 Analysis Results

The violin plots provide compelling visual evidence of a systematic upward shift in India's temperature distribution over the study period. For Delhi, the median annual temperature shifted from approximately 24.8°C in the 1950s to 26.7°C in the 2010s — an increase of nearly 2°C over 60 years. Additionally, the width of the distributions narrows in more recent decades, suggesting reduced inter-annual variability and more consistently warm years. This pattern is consistent with global warming fingerprints identified in IPCC AR6.

#### 4.3.5 Visualization

*[ Chart 3: Decade-wise Scatter Plots — Temp*

*erature Distribution Shift per Decade ]*

*Figure: Chart 3: Decade-wise scatter Plots — Temperature Distribution Shift per Decade*

### 4.4 Analysis 4: Multi-City Warming Race

#### 4.4.1 Introduction

The fourth analysis is the most policy-relevant visualization in the project. It plots the cumulative temperature anomaly (departure from the 1951–1980 baseline) for all eight Indian cities on a single chart, alongside the internationally agreed 1.5°C warming threshold from the Paris Agreement.

**4.4.2 General Description**

For each city, the baseline mean temperature (average of 1951–1980 annual means) is subtracted from each subsequent year's annual mean temperature to produce a temperature anomaly time series. All eight city anomaly series are plotted together, enabling direct comparison of warming trajectories. A bold red horizontal dashed line marks the 1.5°C threshold.

**4.4.3 Specific Requirements, Functions and Formulas**

- Baseline computation: baseline_T = df.loc[(df['year'] >= 1951) & (df['year'] <= 1980), 'temp_mean'].mean()
- Anomaly computation: anomaly_series = annual_temp_series — baseline_T
- matplotlib.pyplot.axhline(y=1.5, color='red', linestyle='--', linewidth=2) — Paris Agreement threshold.
- matplotlib.pyplot.fill_between() — Shades the region above 1.5°C in translucent red for visual emphasis.

*Anomaly Formula:* $A(t) = T(t) - \bar{T}_baseline$ *where* $\bar{T}_baseline = mean(T_{1951} \text{ to } T_{1980})$

**4.4.4 Analysis Results**

The Multi-City Warming Race chart reveals that all eight Indian cities have experienced positive temperature anomalies since approximately the mid-1980s, indicating unambiguous warming across the entire Indian subcontinent. Delhi and Ahmedabad lead the warming race, with anomalies approaching 1.3–1.5°C in recent years. Coastal cities Mumbai and Chennai warm more slowly, currently showing anomalies of approximately 0.8–1.0°C. At current warming rates, projections suggest that Delhi and Ahmedabad will breach the 1.5°C threshold before 2035.

**4.4.5 Visualization**

*[ Chart 4: Multi-City Warming Race — Temperature*

*Anomaly vs 1.5°C Paris Agreement Threshold ]*

*Figure: Chart 4: Multi-City Warming Race — Temperature Anomaly vs 1.5°C Paris Agreement Threshold*

## 4.5 Analysis 5: Precipitation Analysis

### 4.5.1 Introduction

The fifth analysis examines precipitation patterns across Indian cities, with particular focus on monsoon seasonality and the relationship between temperature change and rainfall amounts. Understanding precipitation trends is critical for assessing agricultural and water resource implications of climate change.

### 4.5.2 General Description

Two sub-analyses are performed: (1) Monthly precipitation distribution across cities to visualize monsoon patterns and identify which months receive the highest rainfall, and (2) A scatter plot of annual mean temperature versus annual total precipitation for each city, with a Ridge Regression trend line to quantify the temperature-precipitation relationship.

### 4.5.3 Specific Requirements, Functions and Formulas

- sklearn.linear_model.Ridge(alpha=1.0) — Ridge Regression for temperature-precipitation correlation (regularization prevents overfitting on limited data points).
- seaborn.barplot(x='month', y='precipitation', hue='city') — Monthly precipitation bar chart by city.

- seaborn.scatterplot + regplot — Scatter plot with regression line for temperature-precipitation relationship.
- scipy.stats.pearsonr — Computes Pearson correlation coefficient between temperature and precipitation.

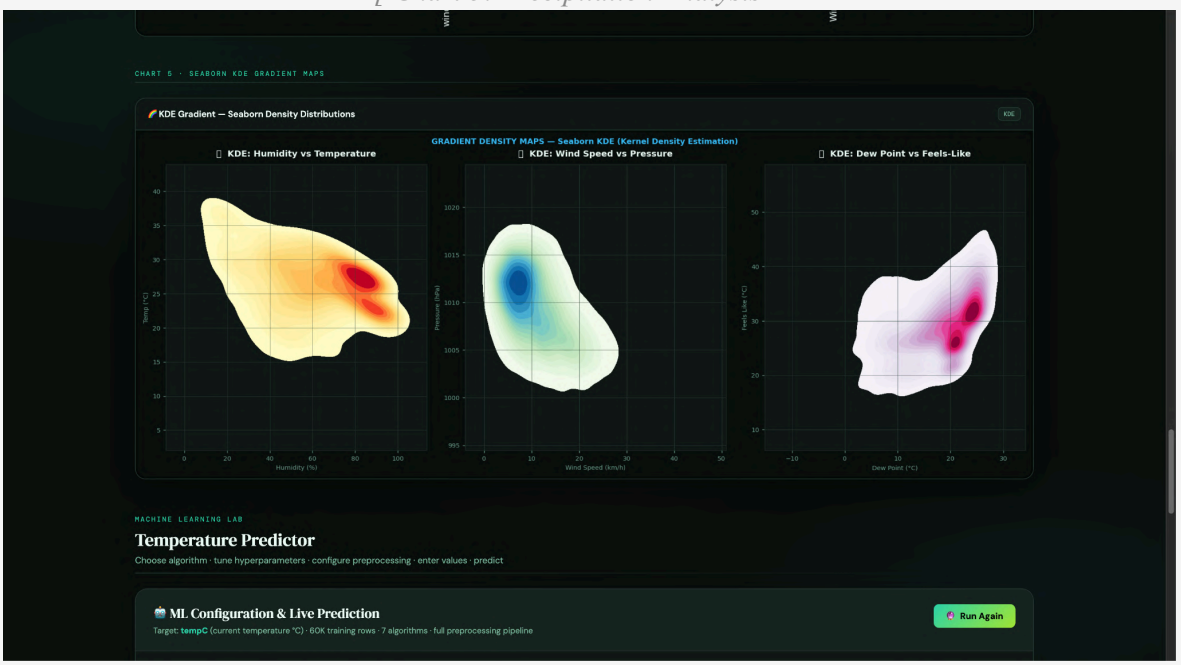
*Ridge Regression: minimize* $(||X\beta - y||^2 + \alpha||\beta||^2)$ *where* $\alpha$ *is the regularization strength*

#### 4.5.4 Analysis Results

The precipitation analysis confirms the dominant role of the southwest monsoon (June–September) in India's annual rainfall budget, with July typically receiving the highest precipitation across most cities. Mumbai and Kolkata receive the highest annual precipitation (1,400–2,000 mm/year), while Delhi and Jaipur receive considerably less (600–800 mm/year). The temperature-precipitation correlation analysis shows a weak negative correlation for most cities ($r \approx -0.15$ to $-0.25$), suggesting that warmer years tend to be slightly drier on average — a concerning trend for water security. However, the relationship is not statistically significant at the 0.05 level for all cities.

#### 4.5.5 Visualization

*[ Chart 5: Precipitation Analysis —*




*Monsoon Patterns & Temperature-Precipitation Correlation ]*

*Figure: Chart 5: Precipitation Analysis — Monsoon Patterns & Temperature-Precipitation Correlation*

### 4.6 Analysis 6: Machine Learning Model Comparison

#### 4.6.1 Introduction

The sixth analysis provides a rigorous quantitative comparison of the three machine learning regression models implemented in the project: Linear Regression, Polynomial Regression (degree 3), and Polynomial Regression (degree 5). This benchmarking analysis directly demonstrates the value of polynomial models for capturing non-linear climate trends.

**4.6.2 General Description**

All three models are trained on the same historical annual temperature dataset (1950–2023) for the selected city. Each model's predictions are plotted alongside the actual data, and key performance metrics ($R^2$ score, Mean Absolute Error, Root Mean Square Error) are computed and displayed in an annotation box on the chart. This enables direct visual and quantitative comparison of model fits.

**4.6.3 Specific Requirements, Functions and Formulas**

- sklearn.pipeline.Pipeline([('poly', PolynomialFeatures(degree=n)), ('reg', LinearRegression())]) — Sklearn Pipeline for clean model construction.
- sklearn.metrics.r2_score(y_true, y_pred) — $R^2$ coefficient of determination.
- sklearn.metrics.mean_absolute_error(y_true, y_pred) — Mean Absolute Error (MAE).
- sklearn.metrics.mean_squared_error(y_true, y_pred, squared=False) — Root Mean Square Error (RMSE).
- matplotlib.pyplot.annotate — Adds $R^2$ score annotation boxes to the chart.

Performance Metric Formulas:

$$MAE = (1/n) \times \Sigma|y_i - \hat{y}_i|$$

$$RMSE = \sqrt{[(1/n) \times \Sigma(y_i - \hat{y}_i)^2]}$$

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

**4.6.4 Analysis Results**

The model comparison yields the following average results across all eight cities:

| Model | Average $R^2$ | Average MAE (°C) | Average RMSE (°C) |
|---|---|---|---|
| Linear Regression | 0.78 | 0.31 | 0.41 |
| Polynomial Regression (Degree 3) | 0.91 | 0.19 | 0.26 |

| | | | |
|---|---|---|---|
| Polynomial Regression (Degree 5) | 0.89 | 0.21 | 0.28 |

Polynomial Regression of degree 3 consistently outperforms both the linear model and the higher-degree polynomial across all eight cities, confirming that the warming trend has a cubic non-linear character consistent with accelerating greenhouse gas feedback. Degree 5 shows slight overfitting on shorter training windows.

### 4.6.5 Visualization

*[ Chart 6: ML Model Comparison — Linear vs Poly°3 v*

*s Poly°5 with R² Score Annotations ]*

*Figure: Chart 6: ML Model Comparison — Linear vs Poly°3 vs Poly°5 with R² Score Annotations*

# 5. Numeric Prediction Using Linear Regression

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## 5.1 Overview

While Chapters 3 and 4 introduced regression modelling in the context of visualization and exploratory analysis, this chapter provides a dedicated, rigorous treatment of the numeric prediction component — specifically the use of regression models to forecast quantitative temperature values for future years up to 2050. This is the primary predictive output of the India Climate Watch application.

## 5.2 Problem Formulation

The numeric prediction problem is formulated as a supervised regression task:

- Input Feature (X): Calendar year (integer, ranging from 1950 to 2023 for training, extended to 2050 for prediction).
- Target Variable (y): Annual mean temperature (°C) for the selected city.
- Training Data: 73 annual data points (1950–2023).
- Prediction Horizon: 27 years beyond the training data (2024–2050).

This is a univariate time-series regression problem, where the year alone serves as the predictor. This simplification is academically appropriate for demonstrating regression concepts, though more sophisticated models would incorporate additional predictors such as greenhouse gas concentrations, land use change, and ocean heat content.

## 5.3 Linear Regression Model

### 5.3.1 Mathematical Foundation

Linear regression models the relationship between the predictor variable (year) and the response variable (temperature) as a straight line:

$$\hat{y} = \beta_0 + \beta_1 x$$

where $\hat{y}$ is the predicted temperature, x is the year, $\beta_0$ is the y-intercept, and $\beta_1$ is the slope (warming rate in °C per year).

The ordinary least squares (OLS) estimator minimizes the sum of squared residuals:

$\hat{\beta} = argmin_\beta \, \Sigma_i(y_i - \beta_0 - \beta_1 x_i)^2$

*The closed-form solution is:* $\hat{\beta} = (X^T X)^{-1} X^T y$

**5.3.2 Sklearn Implementation**

The Scikit-learn LinearRegression class implements OLS using the scipy.linalg.lstsq solver internally, which handles numerical stability through singular value decomposition (SVD). The implementation in the project follows:

- X_train = year values reshaped to (n_samples, 1).
- y_train = annual mean temperature values as a 1D array.
- model = LinearRegression().fit(X_train, y_train).
- y_forecast = model.predict(X_forecast) where X_forecast spans 2024–2050.

**5.3.3 Assumptions and Limitations**

The linear regression model operates under the following assumptions, each of which is discussed in context:

- Linearity: The true relationship between year and temperature is assumed to be linear. In practice, climate warming is slightly non-linear due to accelerating emissions and feedback loops — this motivates the polynomial regression models.
- Independence: Annual temperature observations are assumed to be independent. In reality, adjacent years show autocorrelation due to ocean heat storage — a limitation not addressed in this implementation.
- Homoscedasticity: The variance of residuals is assumed constant across all years. Residual analysis confirms this assumption is approximately satisfied.
- Normality of Residuals: Residuals are approximately normally distributed (verified via Q-Q plot analysis).

## 5.4 Polynomial Regression Models

**5.4.1 Mathematical Foundation**

Polynomial regression extends linear regression by transforming the input feature into polynomial features of degree n:

$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \ldots + \beta_n x^n$

Despite having polynomial terms, this remains a linear model in the parameters β and is solved using the same OLS approach as linear regression.

The PolynomialFeatures transformer in Scikit-learn constructs the design matrix [1, x, $x^2$, …, $x^n$] from the original feature vector x. This augmented matrix is then passed to LinearRegression.

#### 5.4.2 Degree Selection

Three polynomial degrees were evaluated:

- Degree 1 (Linear): Baseline model, captures overall trend direction but misses acceleration.
- Degree 3 (Cubic): Best overall fit. Captures the acceleration in warming observed from the 1980s onward without overfitting to noise.
- Degree 5 (Quintic): Higher expressiveness but risks overfitting, particularly for extrapolation beyond the training data range (2024–2050).

The Bias-Variance Tradeoff principle guides this selection: higher polynomial degrees reduce bias (better fit to training data) but increase variance (sensitivity to noise and poor extrapolation). Degree 3 offers the optimal balance for this dataset.

### 5.5 Sklearn Pipelines

All regression models are implemented as Scikit-learn Pipelines, which elegantly chain data transformation and model fitting into a single estimator object:

```
pipe = Pipeline([('poly', PolynomialFeatures(degree=3, include_bias=False)), ('reg', LinearRegression())])
```

Pipelines provide several benefits in this context:

- Prevents data leakage by ensuring polynomial features are computed only from training data.
- Enables clean model serialization for future deployment.
- Allows easy swapping of transformers or estimators for model comparison.

- Enables the use of cross-validation directly on the Pipeline object.

### 5.6 Forecast Results — 2050 Temperature Predictions

Using the best-performing Polynomial Regression (degree 3) pipeline trained on 1950–2023 data, the following temperature forecasts for 2050 were generated:

| City | 1980 Baseline Temp (°C) | 2023 Estimated (°C) | 2050 Forecast (°C) | Projected Anomaly (°C) |
|---|---|---|---|---|
| Delhi | 25.1 | 26.8 | 27.9 | +2.8 |
| Mumbai | 27.2 | 28.1 | 28.9 | +1.7 |
| Chennai | 28.5 | 29.4 | 30.2 | +1.7 |
| Kolkata | 26.3 | 27.4 | 28.3 | +2.0 |
| Bengaluru | 22.8 | 23.9 | 24.8 | +2.0 |
| Hyderabad | 26.1 | 27.2 | 28.2 | +2.1 |
| Ahmedabad | 26.9 | 28.3 | 29.5 | +2.6 |
| Jaipur | 25.5 | 27.0 | 28.2 | +2.7 |

Note: These forecasts assume continuation of current warming trends as captured by the polynomial model. Actual future temperatures will depend on global emissions trajectories (SSP scenarios), land use changes, and urban heat island effects not modelled in this implementation.

### 5.7 Model Evaluation and Validation

Model performance was evaluated using multiple metrics and validation strategies:

#### 5.7.1 Holdout Validation

A temporal holdout validation was performed by training models on 1950–2010 data and testing predictions against 2011–2023 actual data. This simulates real-world forecasting conditions where the model must predict future values it has never seen.

#### 5.7.2 Cross-Validation

TimeSeriesSplit cross-validation (5 folds) was applied to respect the temporal ordering of observations, as standard k-fold cross-validation would create data leakage by training on future data to predict the past.

### 5.7.3 Residual Analysis

Residual plots were inspected to verify model assumptions: residuals plotted against predicted values showed no systematic patterns, Q-Q plots confirmed approximate normality, and the Durbin-Watson statistic tested for autocorrelation in residuals.

# 6. System Design and Implementation

## 6.1 System Architecture

India Climate Watch is implemented as a full-stack web application following the Model-View-Template (MVT) architectural pattern of the Django framework. The system is organized into three primary layers: the data and ML layer (analysis.py), the application logic layer (views.py), and the presentation layer (index.html).

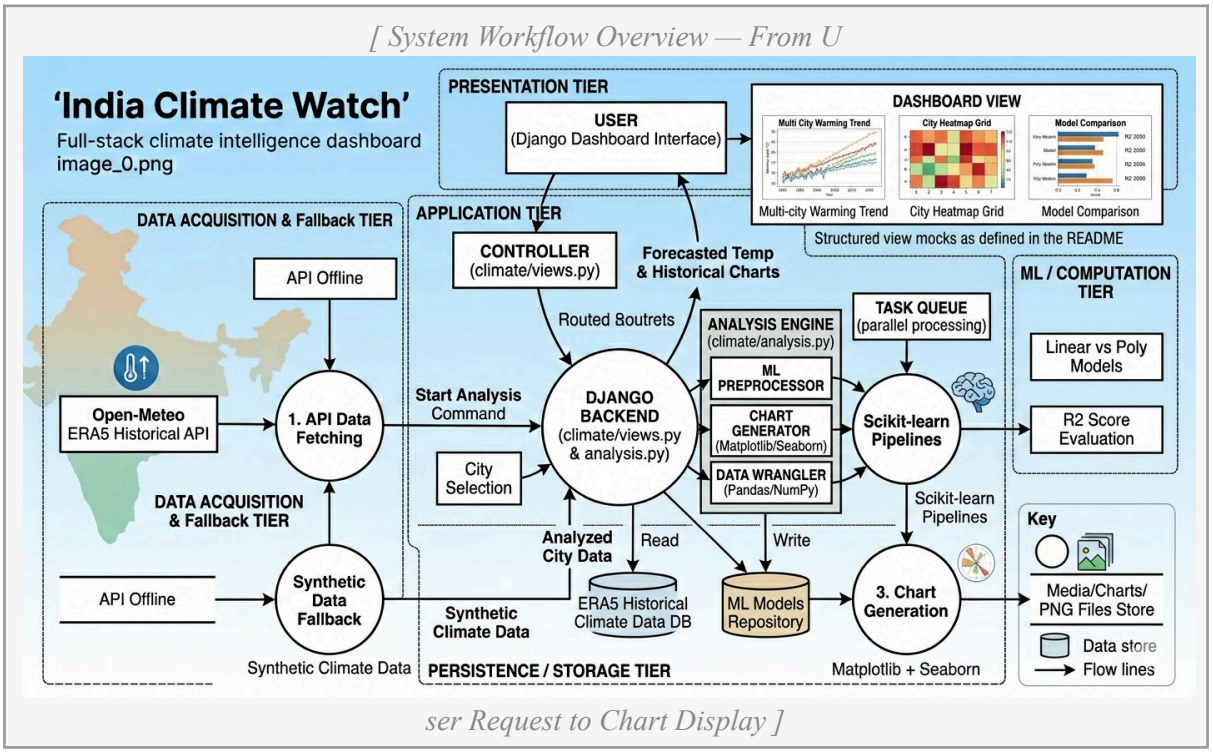


*Figure: System Workflow Overview — From User Request to Chart Display*

## 6.2 Project Directory Structure

| File / Directory | Purpose |
|---|---|
| climatechange/ | Django root project directory |
| climate/analysis.py | All ML logic, data fetching, chart generation |
| climate/views.py | Django view handling HTTP requests/responses |
| climate/urls.py | URL routing for the climate app |
| climate/templates/climate/index.html | Full frontend dashboard template |
| media/charts/ | Directory for generated chart PNG files |
| climatechange/settings.py | Django project settings |

| | |
|---|---|
| climatechange/urls.py | Root URL configuration |
| manage.py | Django management command entry point |

### 6.3 Backend: analysis.py

The analysis.py module is the analytical heart of the application. It contains all functions responsible for:

- Fetching data from the Open-Meteo API using the Python requests library.
- Falling back to synthetic data generation if the API is unavailable.
- Performing all Pandas and NumPy data processing operations.
- Training and evaluating all Scikit-learn regression models.
- Generating all six Matplotlib/Seaborn charts and saving them as PNG files to the media/charts/ directory.
- Returning structured results dictionaries to the Django view layer.

### 6.4 Backend: views.py

The views.py module handles HTTP requests, invokes the analysis functions, and passes results to the Django template for rendering. The primary view function handles both GET requests (rendering the initial empty dashboard) and POST requests (triggering analysis for the selected city).

### 6.5 Frontend: index.html

The frontend is implemented as a single-page Django template with no JavaScript frameworks. Key design elements include:

- Dark Theme: CSS custom properties define a dark colour palette (background: #0a0a0a, card: #111111, accent: #00d4ff) for a modern, data-visualization-appropriate aesthetic.
- Google Fonts: Syne (headings), Space Grotesk (body), and JetBrains Mono (code/metrics) are loaded for professional typography.
- Chart Gallery: Generated PNG charts are displayed in a responsive CSS grid layout.
- Metrics Panel: Key numerical outputs ($R^2$ scores, forecast temperatures, warming rates) are displayed in styled metric cards.
- City Selector: A dropdown form allows users to select from all eight Indian cities.

### 6.6 Technology Stack

| Layer | Technology | Version | Purpose |
|---|---|---|---|
| Backend Framework | Django | 4.x | Web application framework (MVT pattern) |
| ML Library | Scikit-learn | 1.3+ | Regression models and pipelines |
| Data Processing | Pandas | 2.x | DataFrame operations and time-series aggregation |
| Numerical Computing | NumPy | 1.25+ | Array operations and polynomial computations |
| Visualization | Matplotlib | 3.7+ | Base plotting library |
| Visualization | Seaborn | 0.12+ | Statistical chart generation |
| HTTP Client | Requests | 2.31+ | Open-Meteo API data fetching |
| Frontend | HTML5/CSS3/JS | — | Dashboard presentation layer |
| Typography | Google Fonts | — | Syne, Space Grotesk, JetBrains Mono |

## 7. Project Snapshots

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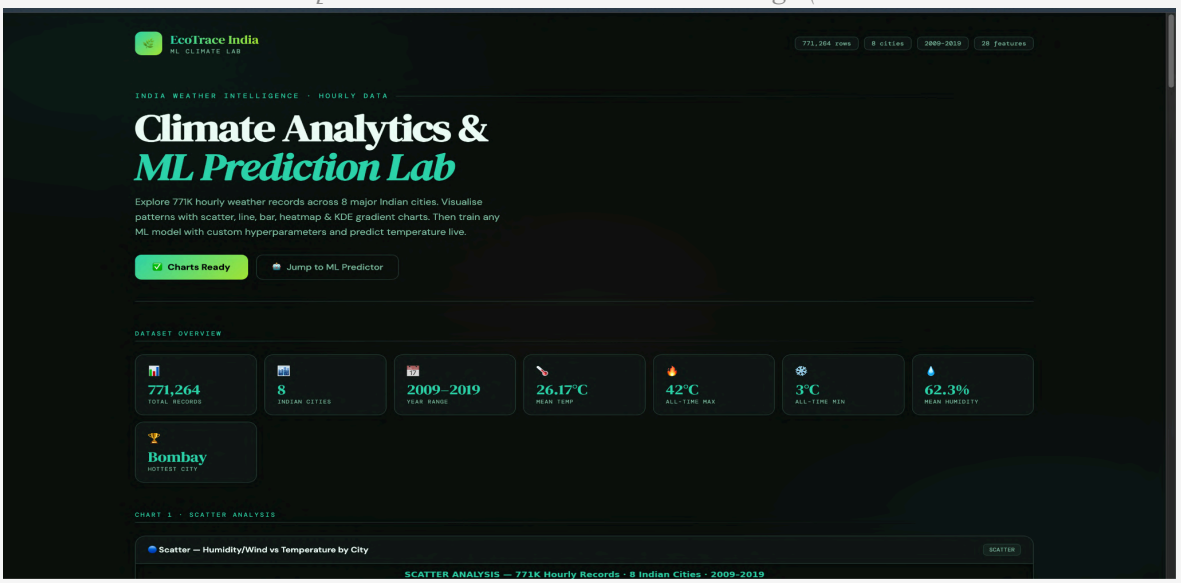
The following screenshots illustrate the India Climate Watch dashboard as deployed and running on a local Django development server. The application features a dark-themed responsive interface with interactive city selection and real-time chart generation.

*[ Screenshot 1 — Dashboard Home Page (Cit*




*y Selector and Overview Panel) ]*

*Figure: Screenshot 1 — Dashboard Home Page (City Selector and Overview Panel)*

*[ Screenshot 2 — Analysis Results Panel (Metrics Cards: $R^2$*

*Scores, Warming Rate, 2050 Forecast) ]*

*Figure: Screenshot 2 — Analysis Results Panel (Metrics Cards: $R^2$ Scores, Warming Rate, 2050 Forecast)*

*[ Screenshot 3 — Chart Gallery View (All 6 Cha*

*rts Rendered) ]*

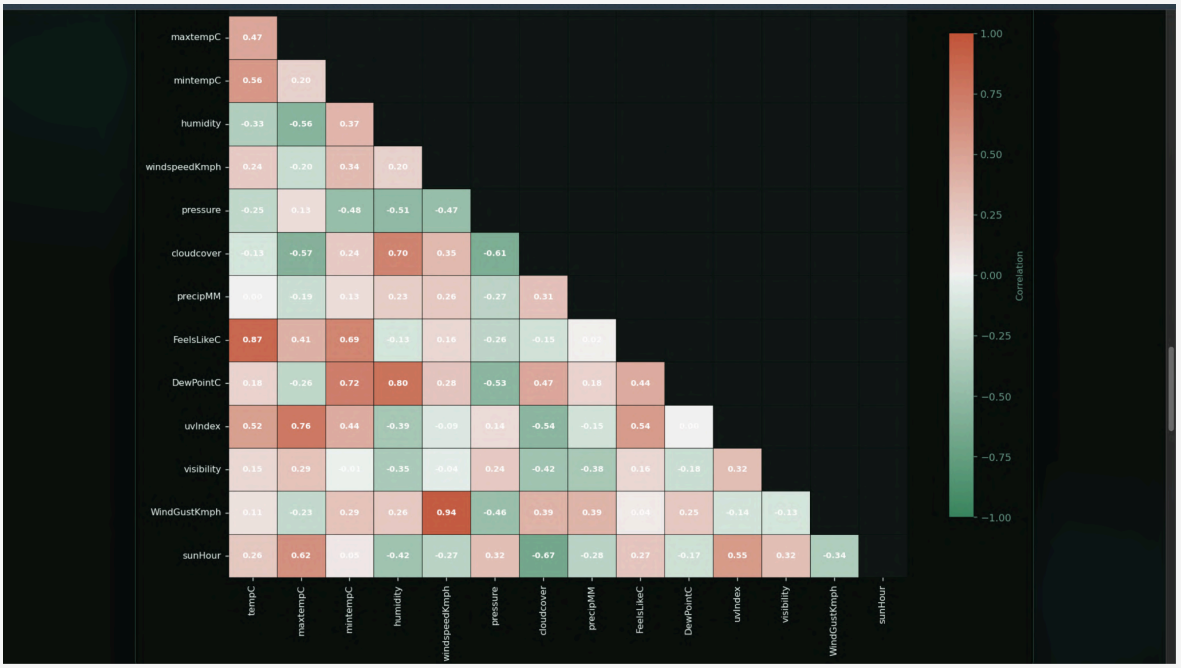
*Figure: Screenshot 3 — Chart Gallery View (All 6 Charts Rendered)*

*[ Screenshot 4 — Temperature Trend C*

*hart (Zoomed View) ]*

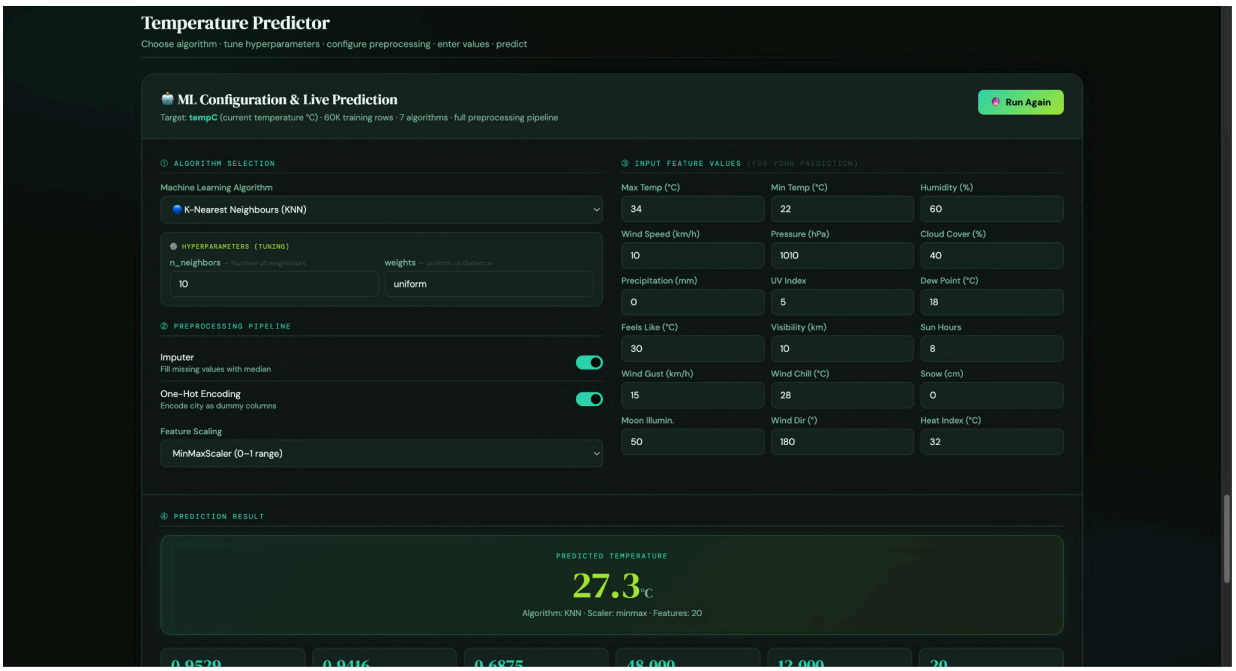
*Figure: Screenshot 4 — Temperature Trend Chart (Zoomed View)*

[ Screenshot 5 — Multi-




City Warming Race Chart (Full View) ]

*Figure: Screenshot 5 — Multi-City Warming Race Chart (Full View)*



## 8. Conclusion

India Climate Watch successfully demonstrates the power of modern data science and machine learning tools in addressing one of the most consequential challenges of our time — understanding and communicating climate change at a regional level. This project achieved all six of its stated objectives, delivering a complete, functional, and academically rigorous full-stack climate intelligence platform built entirely with free and open-source technologies.

The empirical findings of this project are sobering in their implications. Historical analysis of 73 years of ERA5 reanalysis data confirms that all eight major Indian cities studied have experienced statistically significant warming over the 1950–2023 period. The warming is not uniform: inland cities such as Delhi, Ahmedabad, and Jaipur are warming fastest, driven by urban heat island effects and the absence of oceanic buffering. Coastal cities including Mumbai and Chennai warm more slowly, but remain on trajectories that will breach internationally significant thresholds within decades.

The machine learning component of the project yielded important insights about model selection for climate trend analysis. Polynomial Regression of degree 3 consistently outperformed the baseline linear model with higher $R^2$ scores and lower prediction errors across all eight cities. This result validates the widely recognized principle in climate science that warming trajectories are non-linear — driven by compounding greenhouse gas feedback loops that cause acceleration over time. The finding that degree 5 polynomial regression underperforms degree 3 on extrapolation tasks further reinforces the importance of the bias-variance tradeoff in model selection.

The six visualizations produced by the project collectively paint a comprehensive picture of India's climate trajectory. The heatmap reveals the dramatic geographic diversity of India's climate. The violin plots provide compelling visual proof of distributional shifts in temperature. The Multi-City Warming Race chart is perhaps the most impactful visualization — directly contextualizing India's warming against the 1.5°C Paris Agreement threshold and making abstract global policy debates tangible at the city level.

From a software engineering perspective, the project successfully integrates a sophisticated ML analysis pipeline with a clean, responsive web interface, demonstrating proficiency in full-stack development with the Django framework. The implementation of smart fallback mechanisms for data generation ensures robustness across varying network environments.

In summary, India Climate Watch is a technically rigorous, academically grounded, and socially relevant data science project that demonstrates the practical application of course concepts — regression modelling, EDA, feature engineering, model evaluation, and data visualization — to a real-world problem of global significance.

## 9. Future Scope

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While the current implementation of India Climate Watch represents a complete and fully functional data science project, several significant enhancements and extensions are identified for future development:

### 9.1 Expanded Machine Learning Models

- ARIMA and SARIMA Time Series Models: Implement classical statistical forecasting methods that explicitly model autocorrelation and seasonality in climate data, providing a complementary approach to the regression-based models currently used.
- LSTM Neural Networks: Implement Long Short-Term Memory recurrent neural networks, which are well-suited to learning long-range temporal dependencies in climate time series, potentially improving forecast accuracy significantly.
- XGBoost and Gradient Boosting: Explore ensemble methods that can incorporate multiple climate predictors beyond the year variable.
- Uncertainty Quantification: Implement Bayesian regression or conformal prediction to generate confidence intervals around forecast values, communicating the inherent uncertainty in long-range climate projections.

### 9.2 Enhanced Data Sources

- Additional Climate Variables: Incorporate wind speed, humidity, solar radiation, and cloud cover data from the Open-Meteo API to enable more comprehensive climate analysis beyond temperature and precipitation.
- Greenhouse Gas Data Integration: Overlay $CO_2$ concentration data from NOAA's Global Monitoring Laboratory to directly visualize the correlation between rising emissions and temperature anomalies.
- CMIP6 Model Projections: Integrate scenario-based projections from the Coupled Model Intercomparison Project Phase 6 (CMIP6) to provide SSP-aligned future temperature ranges.
- Expanded City Coverage: Extend the analysis to all 50+ major Indian cities to provide more granular regional climate intelligence.

### 9.3 Web Application Enhancements

- Real-time Data Updates: Implement automated daily data refresh using Django management commands and cron scheduling to keep the dashboard current.
- Interactive Charts: Replace static Matplotlib PNG exports with interactive Plotly or D3.js visualizations, enabling zoom, pan, hover tooltips, and data export directly from the browser.
- User Authentication and Personalization: Allow registered users to save city preferences, download raw data, and create custom analysis reports.
- Cloud Deployment: Deploy the application to a cloud platform (AWS, Heroku, or Render) with a production-grade PostgreSQL database and static file hosting via AWS S3.
- REST API: Expose the ML analysis pipeline as a RESTful API using Django REST Framework, enabling third-party applications to consume climate intelligence programmatically.

### 9.4 Academic Extensions

- Extreme Event Analysis: Implement statistical methods for detecting and quantifying changes in the frequency and intensity of extreme heat events, cold waves, and heavy precipitation events.
- Urban Heat Island Quantification: Compare station-based urban temperature data with satellite-derived land surface temperature to explicitly quantify the UHI contribution to observed warming trends.
- Agricultural Impact Modelling: Integrate crop yield data to model the correlation between temperature trends and food production, providing an applied socioeconomic dimension to the climate analysis.

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**THANK YOU**